

CAMBRIDGE 9709 - PURE MATHEMATICS 1

Final P1 mock

Full syllabus checkpoint - 75 marks

YOUR AIM TODAY

Complete a timed 1 hour 50 minute paper across all eight P1 sections.

YOUR RULE

Closed book. Calculator allowed. Show sufficient working. Give exact answers unless instructed otherwise. Stop at 110 minutes.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

INSTRUCTIONS

Candidate information

Ten structured questions. Answer every question.

Time	1 hour 50 minutes
Marks	75
Questions	10 compulsory structured questions
Coverage	All eight Pure Mathematics 1 sections

TIMING

Work at roughly 1.3 minutes per mark and reserve the final 10 minutes for checking.

Quadratics and functions

QUESTION 1

Quadratic structure

[7 marks]

(a) Express $2x^2-12x+11$ in completed-square form. [3]

(b) Hence solve $2x^2-12x+11=0$ exactly. [2]

(c) Solve $2x^2-12x+11>0$. [2]

Functions

QUESTION 2

Functions

[7 marks]

Let $f(x)=3x+1$ and $g(x)=x^2-4$ with domain $x \geq 0$. [0]

(a) Find $fg(x)$. [2]

(b) Find $gf(x)$. [2]

(c) Find $g^{-1}(x)$. [2]

(d) State the range of g . [1]

Coordinate geometry and circular measure

QUESTION 3

Straight-line geometry

[8 marks]

Points $A=(-1,2)$ and $B=(5,10)$. Find the gradient of AB. [2]

(b) Find the exact length AB. [2]

(c) Find an equation of AB. [2]

(d) Find the perpendicular bisector of AB. [2]

QUESTION 4

Circular measure

QUESTION 4

Sector and segment

[7 marks]

A sector has radius 6 cm and angle $2\pi/3$ radians. Find its arc length. [2]

(b) Find its area. [2]

(c) Find the area of the minor segment. [3]

Trigonometry

QUESTION 5

Exact values and transformations

[7 marks]

(a) State the exact values of $\sin(7\pi/6)$ and $\tan(3\pi/4)$. [2]

For $y=2\sin(3x)-1$, state (b) the amplitude and period. [2]

(c) State the range. [1]

(d) Solve $\cos x = -\sqrt{3}/2$ for $0 \leq x \leq 2\pi$. [2]

QUESTION 6

Trigonometric identities and equations

QUESTION 6

Identities and equations

[8 marks]

(a) Prove $(1 - \cos^2 x)/\sin x$ is identical to $\sin x$. [2]

(b) Solve $2\sin^2 x - 3\sin x + 1 = 0$ for $0 \leq x \leq 360$ degrees. [4]

(c) Solve $\tan(2x) = 1$ for $0 \leq x \leq 180$ degrees. [2]

Series

QUESTION 7

Progressions and binomial expansion

[8 marks]

An AP has third term 11 and eighth term 31. Find a and d . [3]

(b) Find the sum of its first 20 terms. [2]

(c) Find the coefficient of x^3 in $(2-x)^6$. [3]

Differentiation

QUESTION 8

Stationary points

[8 marks]

Given $y=2x^3-9x^2+12x$, find dy/dx and factorise it. [2]

(b) Find the coordinates of both stationary points. [3]

(c) Classify both points. [2]

(d) Find the tangent at $x=0$. [1]

Integration and mixed calculus

QUESTION 9

Integration

[8 marks]

(a) Integrate $6x^2-4x+3$. [2]

Given $dy/dx=6x^2-4x+3$ and $y=5$ at $x=1$, find y in terms of x . [2]

(c) Evaluate the integral from 0 to 2 of $6x^2-4x+3$. [2]

(d) The region under $y=\sqrt{x}$, from $x=0$ to $x=4$, rotates about the x -axis. Find its volume. [2]

Curve analysis

QUESTION 10

Curve analysis

[7 marks]

The curve is $y=4x-x^2$. Find its x-axis intersections. [1]

(b) Find the coordinates and nature of its stationary point. [3]

(c) Find the area between the curve and the x-axis. [3]

END OF PAPER

Stop at 110 minutes. Record the score by topic in the tracker before opening the mark scheme.